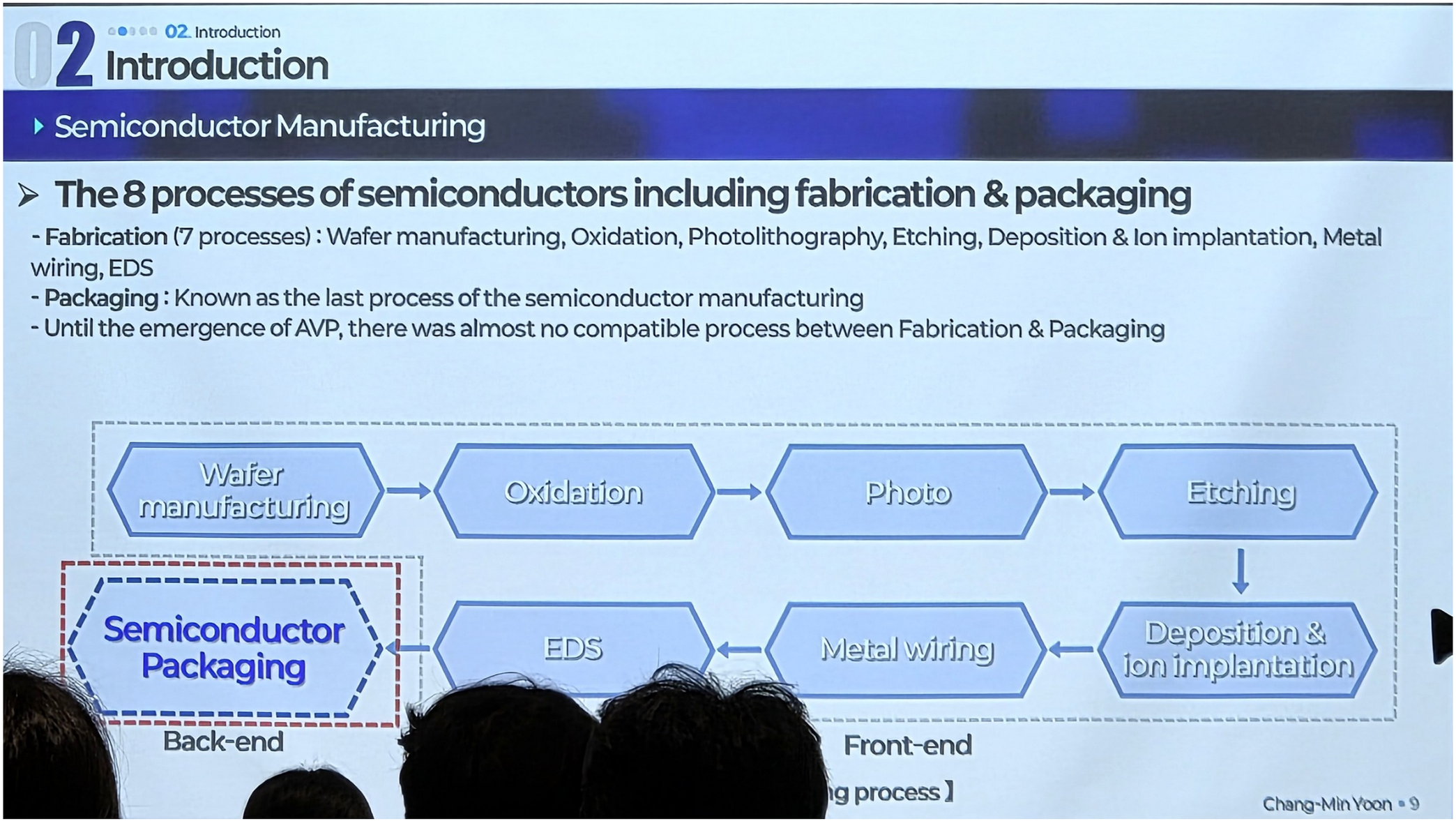
T36 저온 하이브리드 본딩과 고분자 계면·언더필 설계

Inha University · 윤창민 · Chang-Min Yoon



02 Introduction

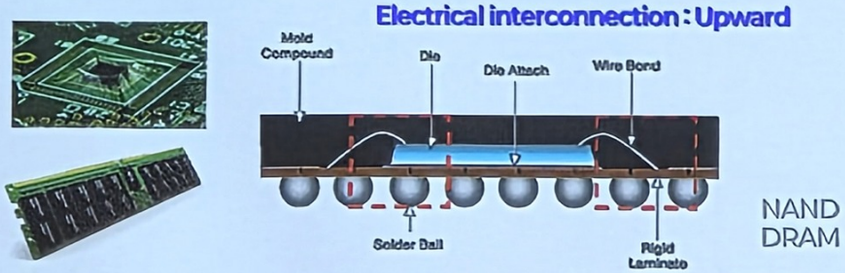
02 Introduction

Semiconductor Packaging

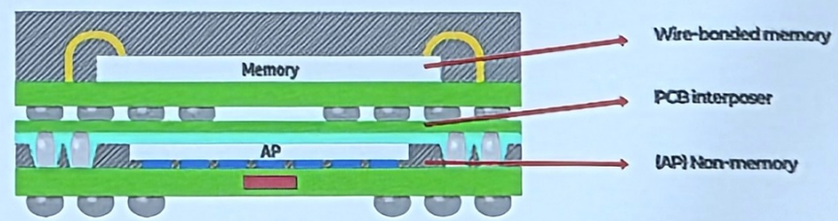
Conventional packaging

* PoP = Package-on-Package

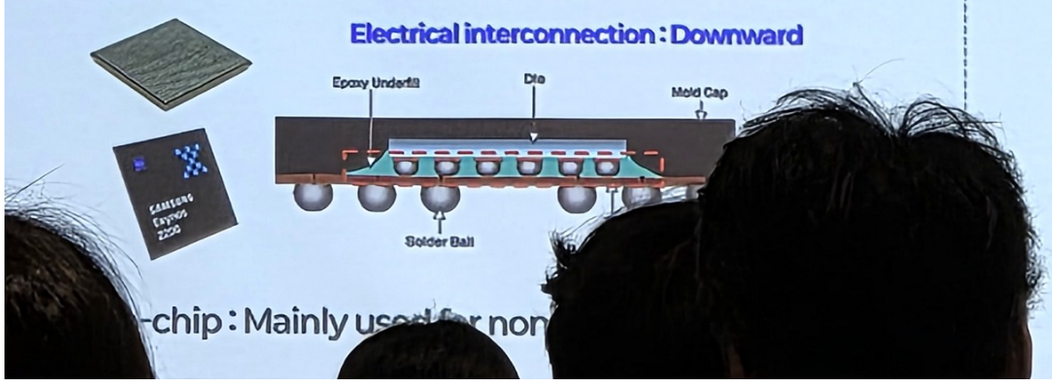
- Convenional packaging uses **PCB as a substrate for electrical interconnection (Accounts for 60% of the market)**
- Interconnection : Memory = **Wire Bonding**, Non-Memory = **Solder bump**



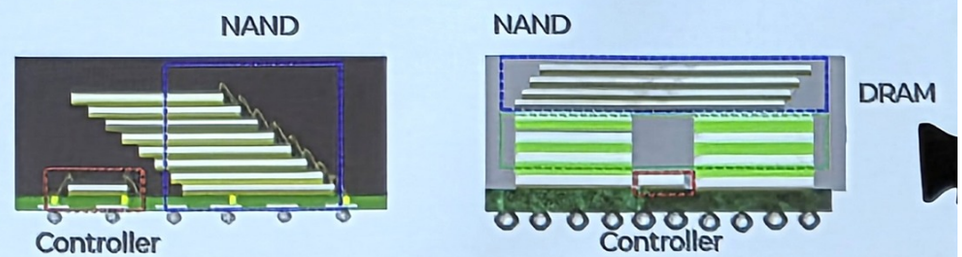
【 Wire bonding : Mainly used for memory devices 】



【 Package-on-package (ePoP) : Non-memory + memory 】



-chip : Mainly used for non



eMMC (8N+1C)

eMCP (N+C+D)

【 Memory + Non-memory form factors : PoP/eMMC/eMCP 】

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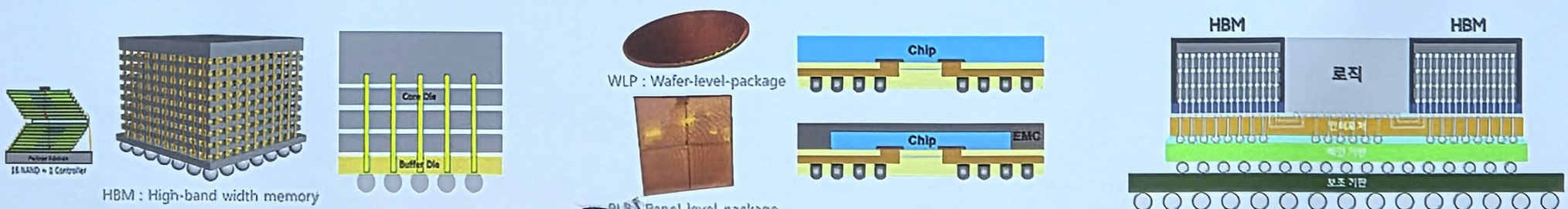
Semiconductor Packaging

Advanced packaging

- Key technologies : Heterogeneous integration, RDL substrates (PCB X), Front-end-level technologies, Silicon & Glass interposers
- Inteconnection : micro solder bump, hybrid bonding (without bump, but direct chip-to-chip connection)
- Key Product : HBM (memory), WLP/PLP (logic), System-in-package (Heterogeneous integration)

☞ Heterogeneous integration is the core of advanced packaging, enabling different types of chips to be closely connected and operate as a single high-performance system.



【High-band width (HBM)】 【Non-memory devices】 【System-in-package】

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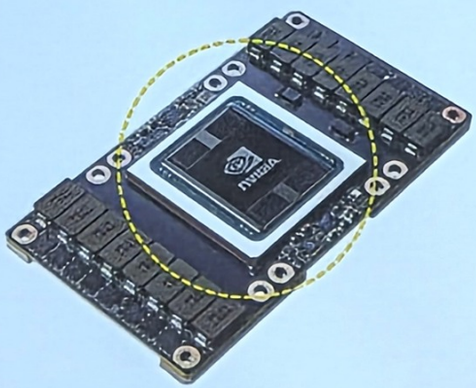
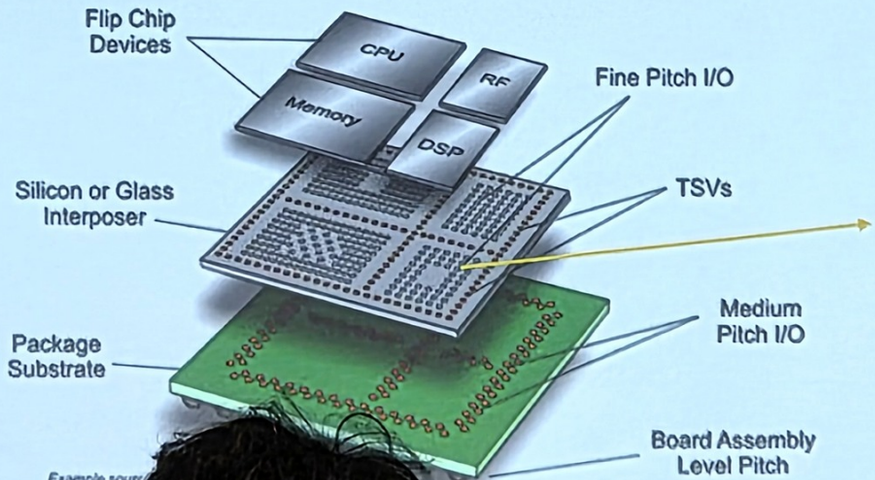
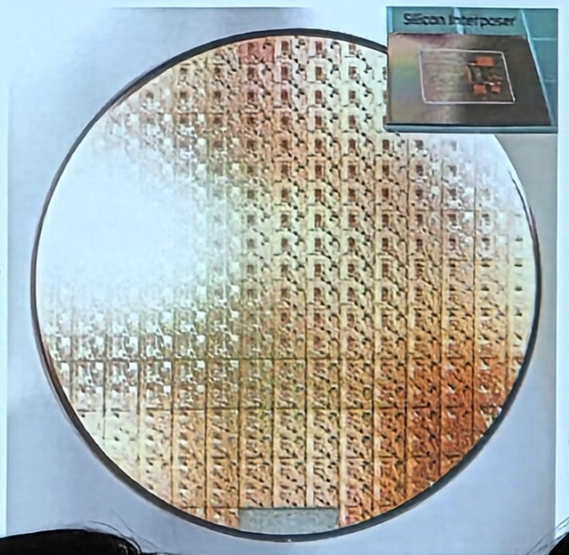
Semiconductor Packaging

➢ Advanced packaging : System-in-Package (SiP) utilizing silicon interposer

- System-in-Package (SiP), one of the most promising next-generation packaging technologies, integrates different types of semiconductor chips into a single system (**Heterogeneous integration**)
- **Various chips are assembled on an interposer, allowing them to operate as a single system**







ge architecture 】

nterposer 】

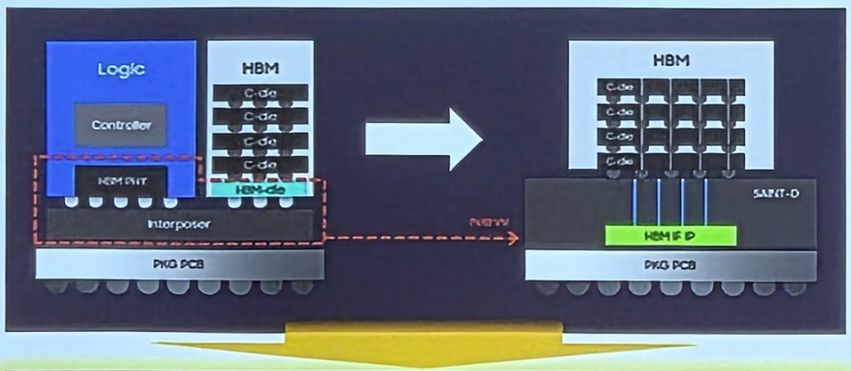
02 Introduction
02 Research Agenda

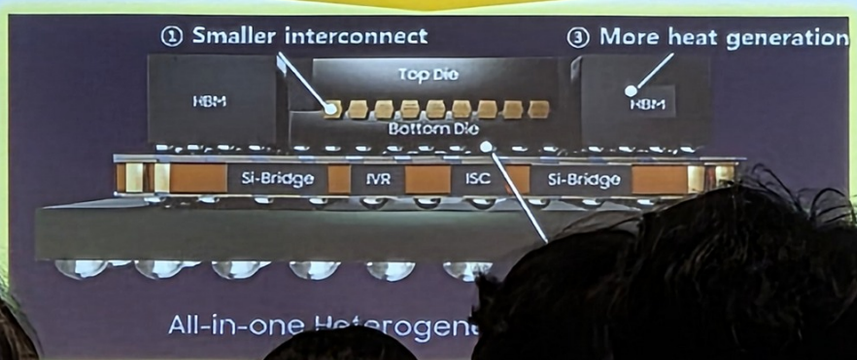
Advanced Packaging and Heterogeneous Integration : Materials and Interface Bottlenecks

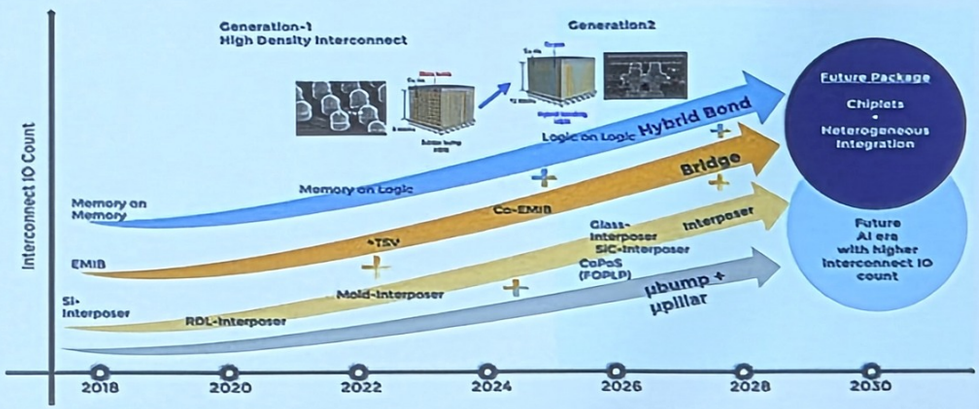
➢ Advanced Packaging and Heterogeneous Integration

☞ Materials and Interface Bottlenecks (3 keywords)

① **Smaller interconnect pitch**: fine-pitch bonding bottleneck
② **Larger heterogeneous area**: warpage and stress bottleneck
③ **Higher vertical stack**: thermal & warpage bottleneck







Interposer size	<2500 mm2	>3400 mm2
pitch	40 – 20 um	<20 um
size	40 – 20 um	<20 um

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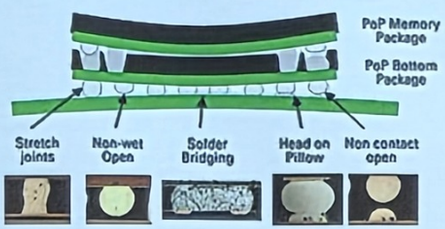
02 Research Agenda

▸ Advanced Packaging and Heterogeneous Integration : Materials and Interface Bottlenecks

➢ Warpage and Mechanical Reliability : Stress control in heterogenous stacks

① CTE and modulus mismatch among Si, Cu, substrate, EMC, and underfill
② Polymer formulation for stress relaxation and warpage suppression
③ Residual stress accumulation in large-area and multi-die packages

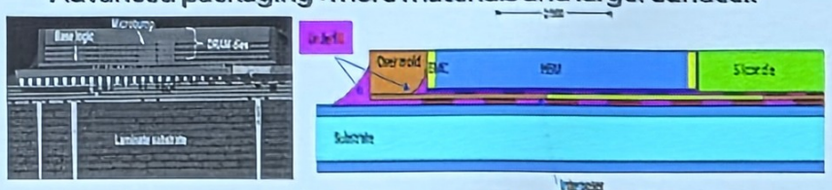
· Factors for warpage : Difference in CTE and modulus of materials



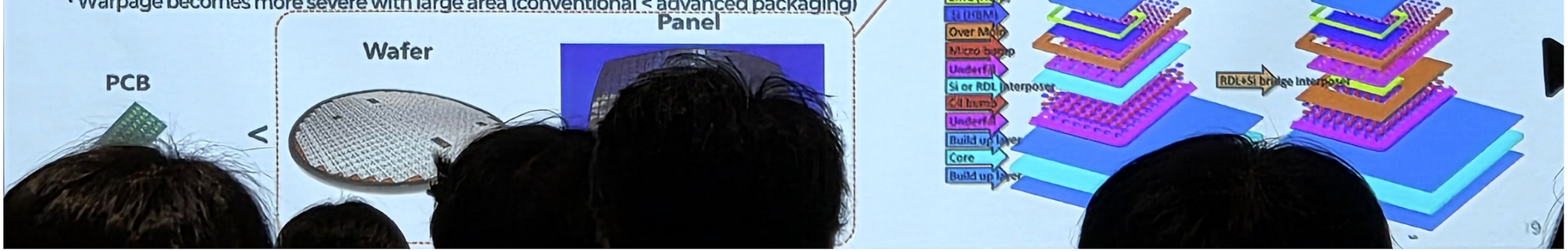
Material	Young's modulus GPa	Poisson's ratio	CTE 10^{-6} ℃$^{-1}$
Sn3.0Ag0.5Cu	38.7	0.35	25
Copper pad	120	0.345	17
Solder mask	5	0.3	95
BT substrate	26	0.39	15
PCB board	22	0.28	18
Mold compound	10	0.25	14.1
Si die	131	0.25	2.8

Note: CTE—coefficient of thermal expansion

· Advanced packaging : more materials and larger surface!!



· Warpage becomes more severe with large area (conventional < advanced packaging)



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Advanced Packaging and Heterogeneous Integration : Materials and Interface Bottlenecks

Thermal management : Heat dissipation across materials and interfaces

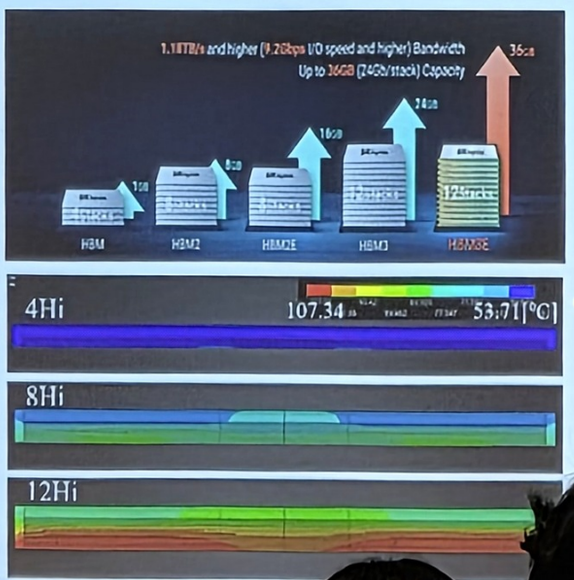
① High-k filler selection: ceramic, nitride, and carbonaceous fillers

② Thermal pathway formation through filler network design

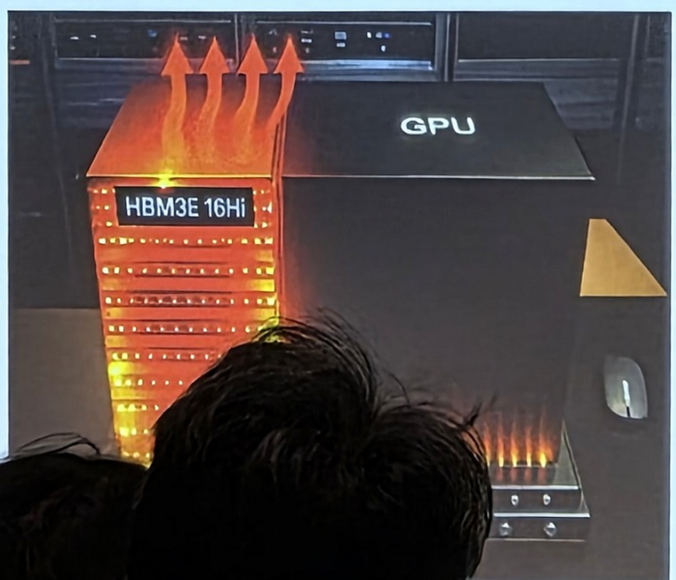
③ Interface control for reducing thermal resistance

- Next generation material (thermal managing)
- EMC : > 3.0 W, Underfill > 2.0 W

·12-hi stacks already facing thermal problem



· Importance of thermal management in increasing!!



· Materials with high thermal conductance is required!!

	~2023	2024	2025	2026	2027	2028	2029	2030
PKG Structure	2.5D		2.5D, RDL interposer, Embedded, 3D					
Interposer Size	50 x 50 mm		65 x 75 mm			> 80 x 80 mm		
Interposer Process	Wafer					Wafer/Panel		
Substrate Size	110 x 110 mm		120 x 120 mm			> 150 x 150mm		
Power Consumption	~1000 W		~1500 W			> 2000 W		
TIM (TR Index)	TC-CWP04		TC-CWP04(60um)		Next TC-2		Next Gen.	
	100 (Ref.)		70			35		20
EMC (Thermal Conductivity) Granule	GE-S06HT series		High K Type			Advanced High K		
	3 W, for FCCSP		3 W, for WL/PL			5 W, for WL/PL		
Film	EB-0005		High K Type			Advanced High K		
	1 W		3 W					
CUF (Thermal Conductivity)	CEL-C-3720-4HT45R		3780T-6A			Next Gen.		
	0.5 W		2.0 W			> 2.0 W		
DAF (Therm			HR-150 series			Next		
			1.5 W			1.5~2.0 W		

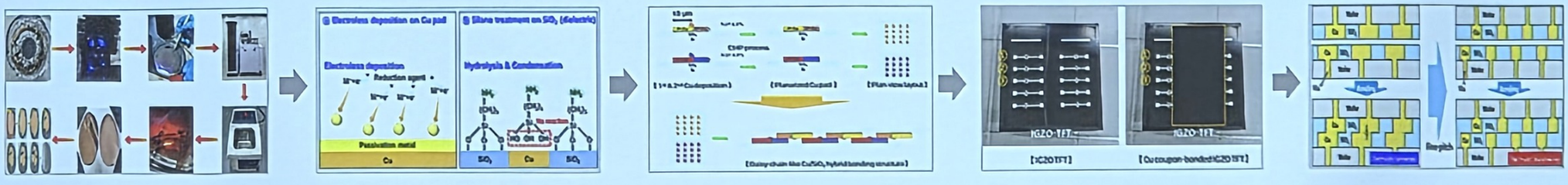
03. Interconnection

03 Interconnection

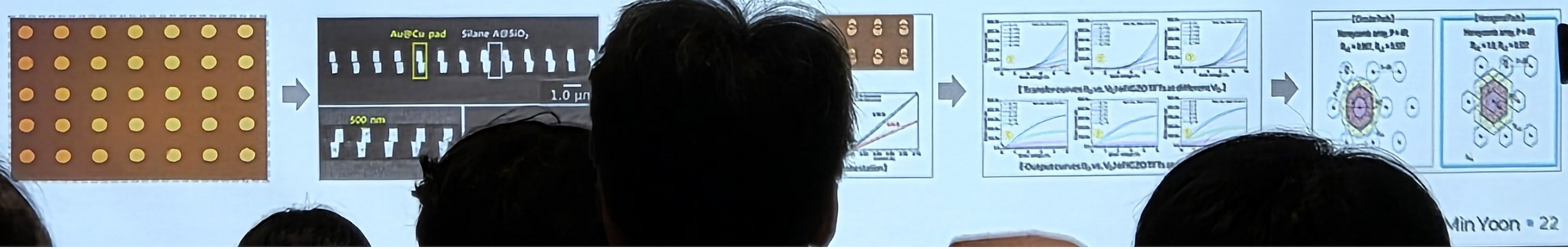
Interconnection : Fine-pitch and low-temperature bonding

Research summary

- We aim to advance hybrid bonding from micron-scale test vehicles to sub-micron interconnections, electrically validated bonding pads, transistor-integrated structures, and alignment-tolerant pad designs



Micron-pitch → Sub-micron pitch → Electrical validation → Transistor integration → Pad design



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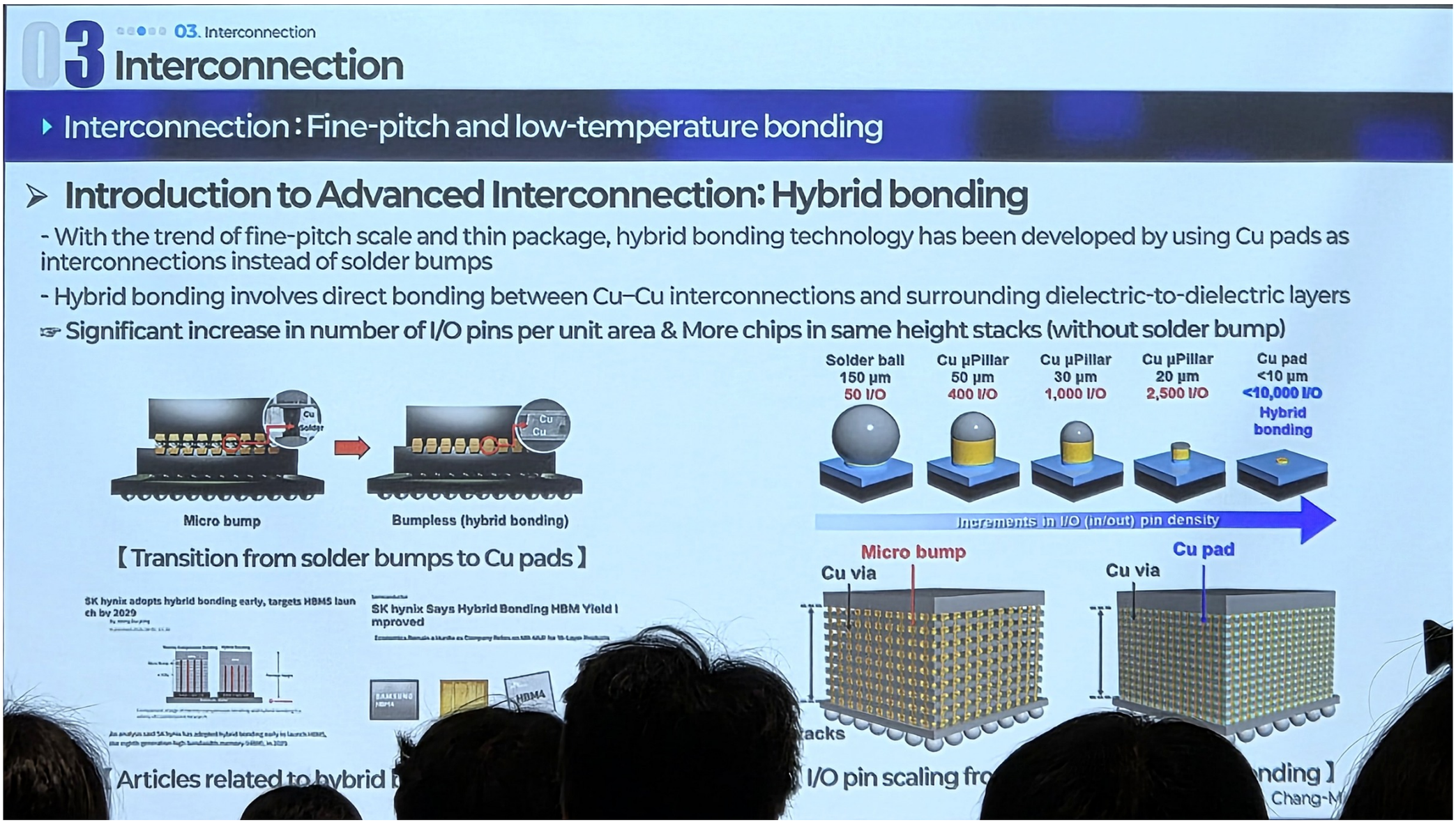
03 Interconnection
03. Interconnection
Interconnection : Fine-pitch and low-temperature bonding
Introduction to Advanced Interconnection: Hybrid bonding
- With the trend of fine-pitch scale and thin package, hybrid bonding technology has been developed by using Cu pads as interconnections instead of solder bumps
- Hybrid bonding involves direct bonding between Cu–Cu interconnections and surrounding dielectric-to-dielectric layers
☞ Significant increase in number of I/O pins per unit area & More chips in same height stacks (without solder bump)
Micro bump
Bumpless (hybrid bonding)
【Transition from solder bumps to Cu pads】
Solder ball 150 μm 50 I/O
Cu μPillar 50 μm 400 I/O
Cu μPillar 30 μm 1,000 I/O
Cu μPillar 20 μm 2,500 I/O
Cu pad <10 μm <10,000 I/O Hybrid bonding
Increments in I/O (in/out) pin density
Micro bump
Cu via
Cu pad
Cu via
SK hynix adopts hybrid bonding early, targets HBM5 launch by 2029
SK hynix Says Hybrid Bonding HBM Yield Improved
Articles related to hybrid
I/O pin scaling fro
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03 Interconnection

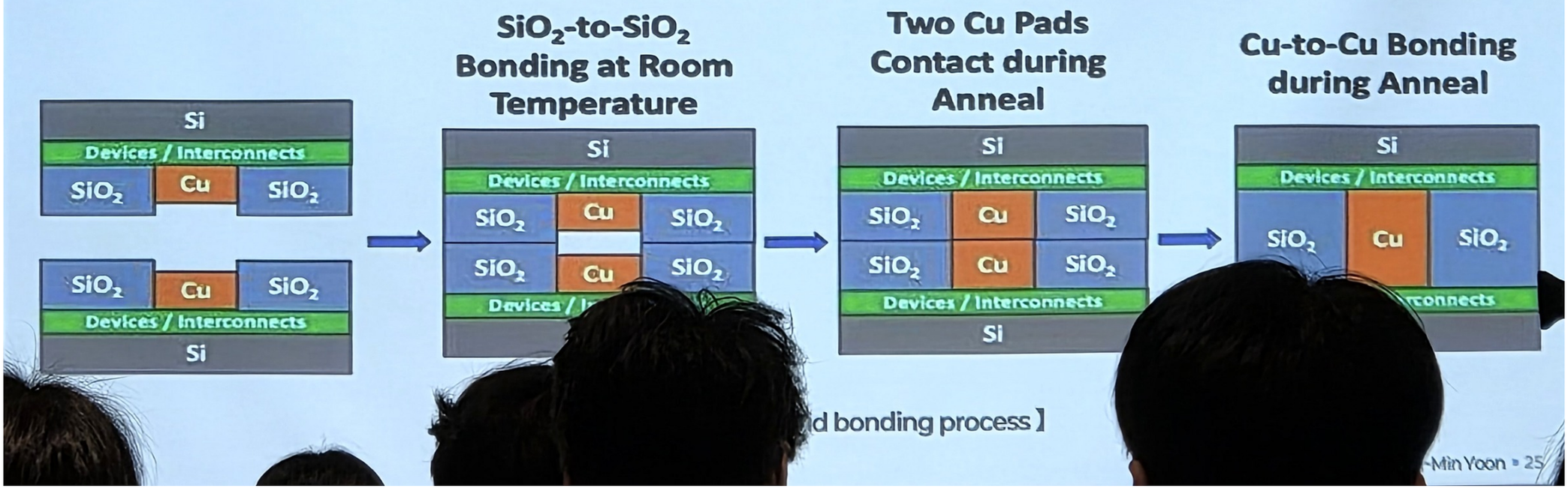
03 Interconnection

Interconnection : Fine-pitch and low-temperature bonding

Commercial standard interface for hybrid bonding: Cu/SiO_2

- The currently reported standard bonding interface for hybrid bonding is Cu/SiO_2 (Hybrid interface - Metal : Cu, Dielectric : SiO_2)
- Hybrid bonding process flow:

Chip-to-chip bonding → Physical contact of SiO_2–SiO_2 → Annealing (Cu–Cu metal diffusion and SiO_2–SiO_2 chemical bonding)



03. Interconnection

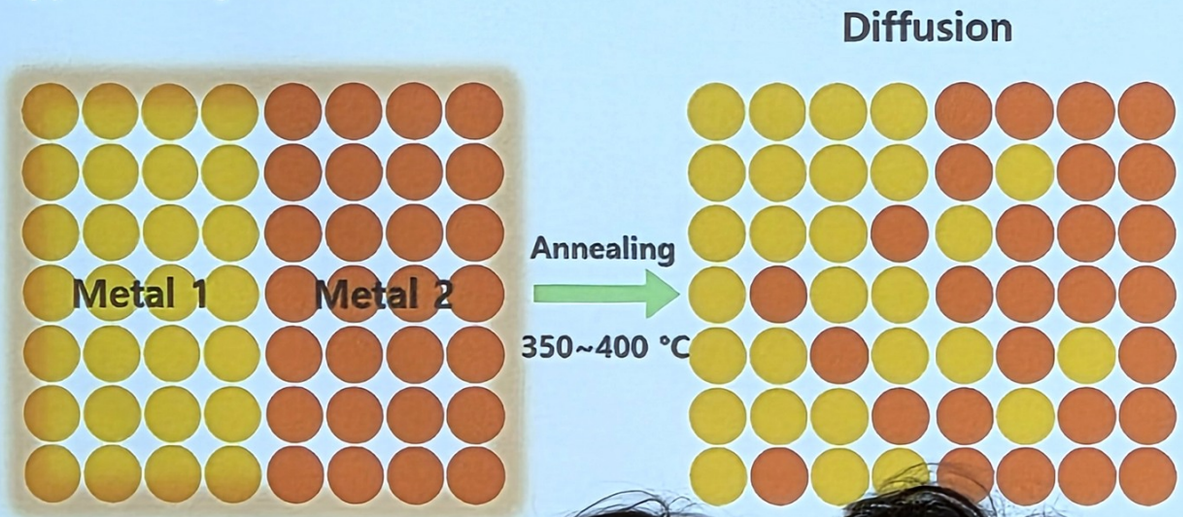
03 Interconnection

Interconnection : Fine-pitch and low-temperature bonding

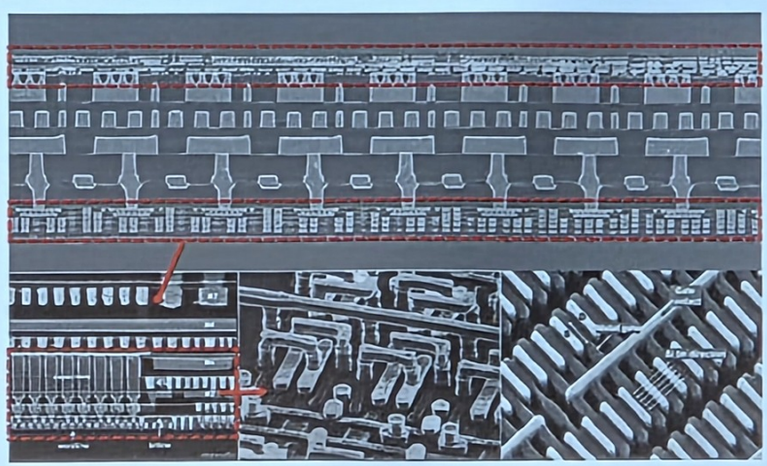
➢ Technical challenge of hybrid bonding : Process temperature

- Cu–Cu bonding requires a high process temperature of 350~400 °C (Cu diffusion temperature)
- Temperatures above 300 °C can damage highly integrated semiconductor devices and transistors

☞ Cu–Cu and dielectric–dielectric bonding must be achieved at optimized semiconductor packaging process temperature (approximately 250 °C)



【Me… …】



【Highly integrated devices and transistors】

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03. Interconnection

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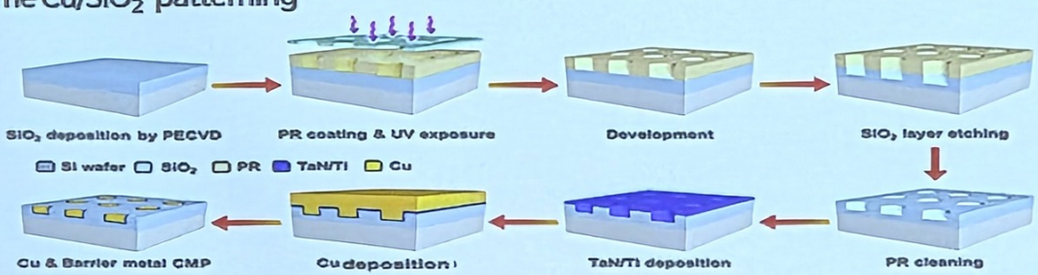
Interconnection : Fine-pitch and low-temperature bonding

Micron-pitch Cu/SiO_2 hybrid bonding – In-house PKG Fab.

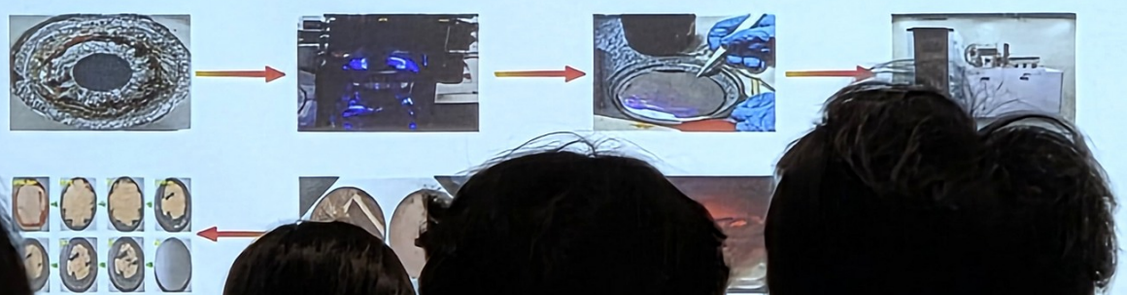
- In-house package fabrication = mask aligner, thermal evaporator, and etching instrument
- Cu/SiO_2 hybrid bonding specimen with Cu pad diameter of 10 μm (25 μm pitch) on wafer was successfully synthesized
- This platform enables practical evaluation of heterogeneous Cu/SiO_2 hybrid bonding

【 Scheme & practical process of in-house PKG Fabrication 】

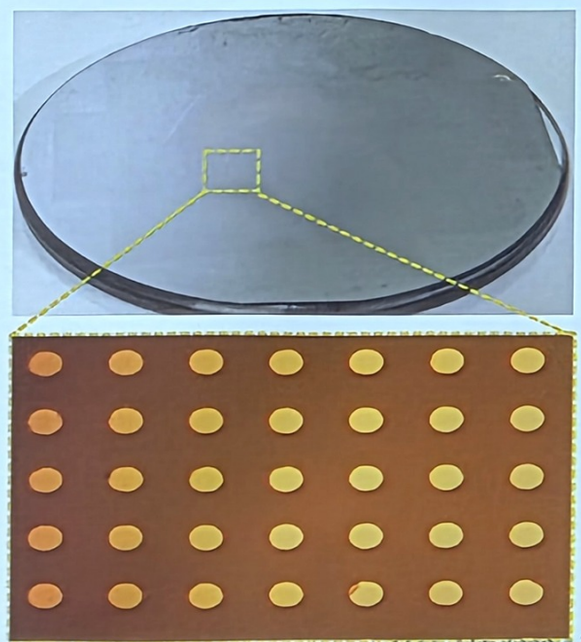
* Scheme Cu/SiO_2-patterning



* Practical process of in-house PKG Fab



【 Cu/SiO_2-patterned wafer 】



03. Interconnection

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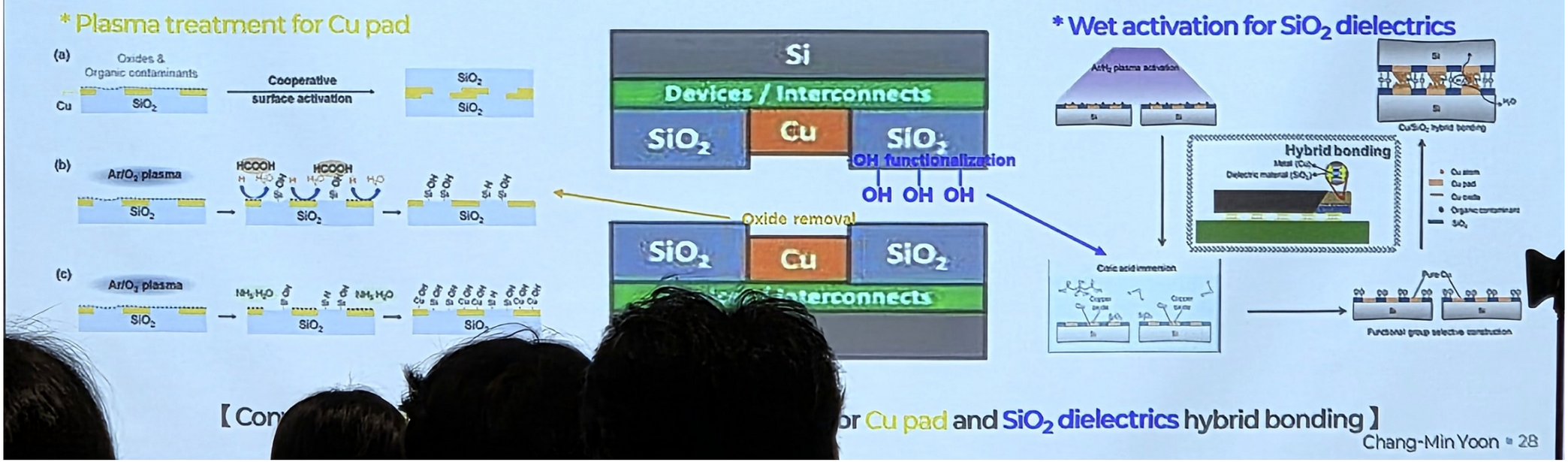
Interconnection : Fine-pitch and low-temperature bonding

➢ Micron-pitch Cu/SiO_2 hybrid bonding – Conventional surface activation

- Plasma treatments such as Ar, O_2, N_2, and H_2 have been widely used for Cu cleaning, oxide removal, and surface activation.
- Wet chemical (organic acid & base) treatments, have been introduced to functionalize SiO_2 dielectric surfaces.
- However, conventional activation approaches are limited in usage of harsh conditions; plasma treatment & acid/base usage

☞ Thus, mild and selective interface engineering is necessary for fine-pitch hybrid bonding as interconnect pitch continues to shrink



【Con or Cu pad and SiO_2 dielectrics hybrid bonding】

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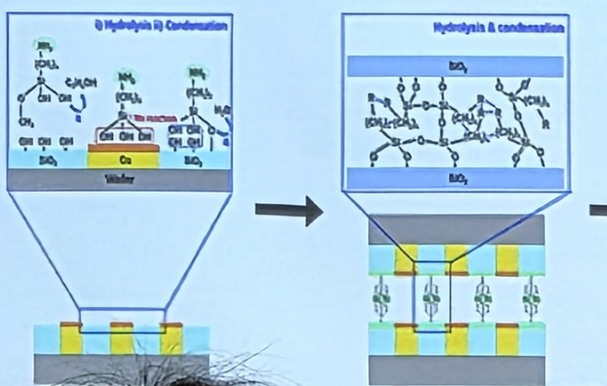
03. Interconnection

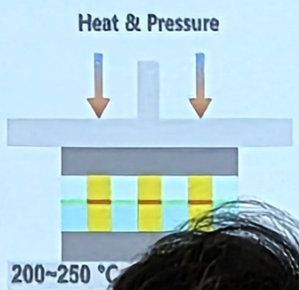
03 Interconnection

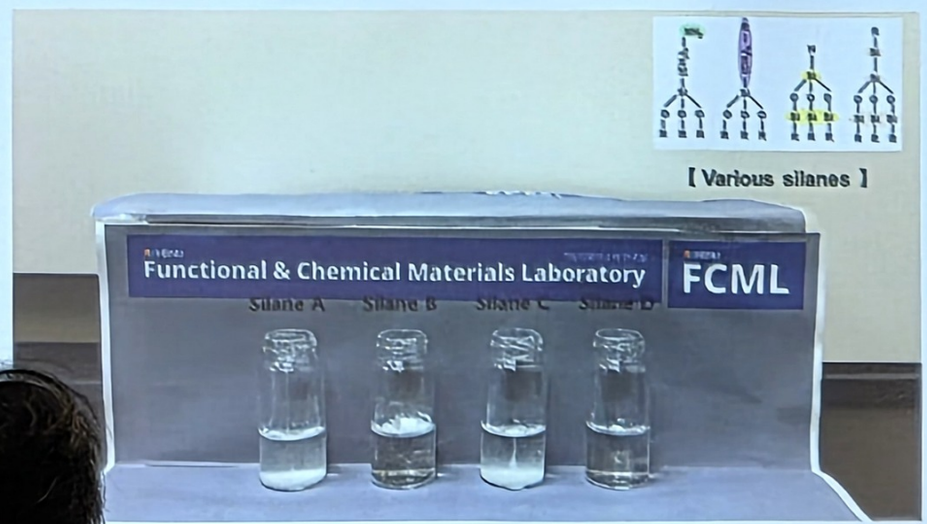
Interconnection : Fine-pitch and low-temperature bonding

➢ Micron-pitch Cu/SiO_2 hybrid bonding – Selective silane treatment on dielectric

- **Silane treatment is a facile & mild strategy for selectively coating of silane precursors onto SiO_2 dielectric layer** through hydrolysis and condensation reactions (spin coating method) **within a short processing time (< 1 min)**
- Depending on the functional groups of silane tails, the bonding temperature can be reduced by enhancing interfacial adhesion







【 Me ectric layer for hybrid bonding 】

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03 Interconnection

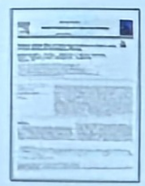
03 Interconnection

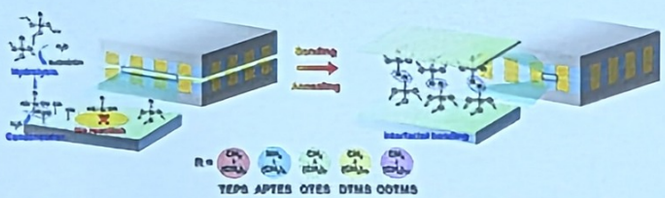
Suk Jekal
Gyu-Sik Park
Seong Uk Kang

Interconnection : Fine-pitch and low-temperature bonding

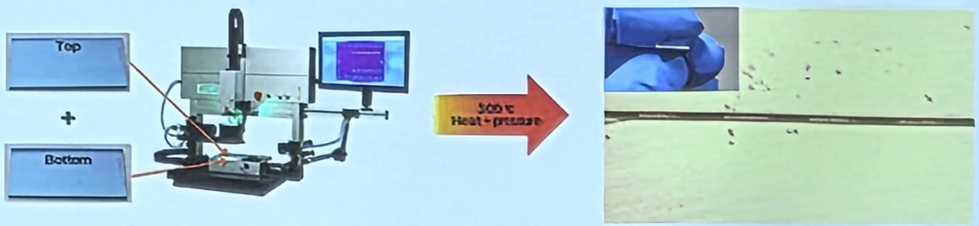
➢ Micron-pitch Cu/SiO_2 hybrid bonding – Silane treatment on dielectric

- Silane treatment for SiO_2 dielectric with short processing time
- By increasing the adhesion of SiO_2 dielectrics, the bonding temperature was successfully lowered to 300 °C

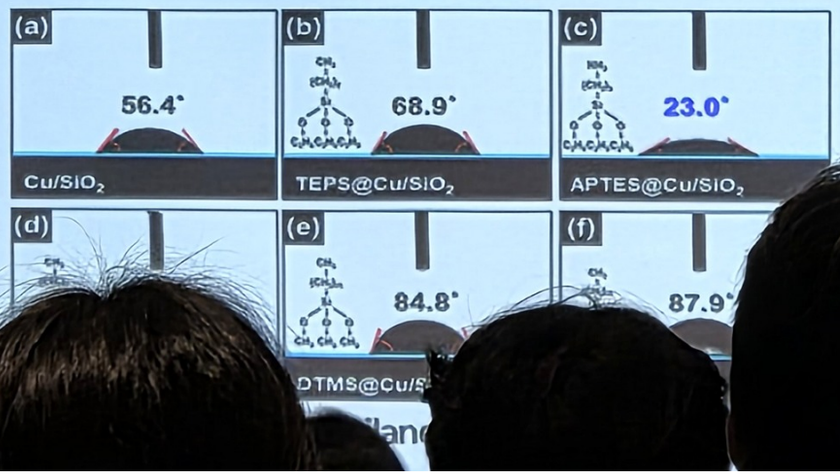


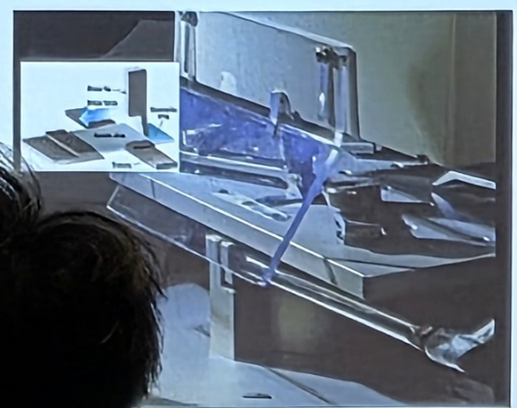


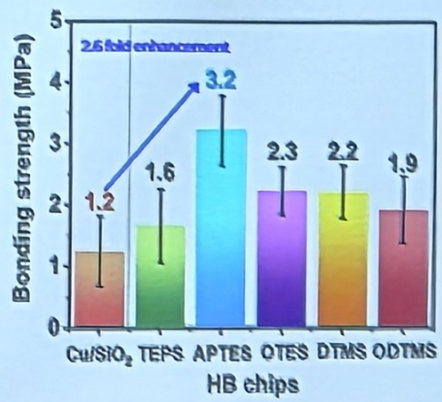
【Schematic illustration of selective silane treatment on SiO_2】



【Chip bonder : Low-temperature hybrid bonding】







【Die shear test of bonded chips】

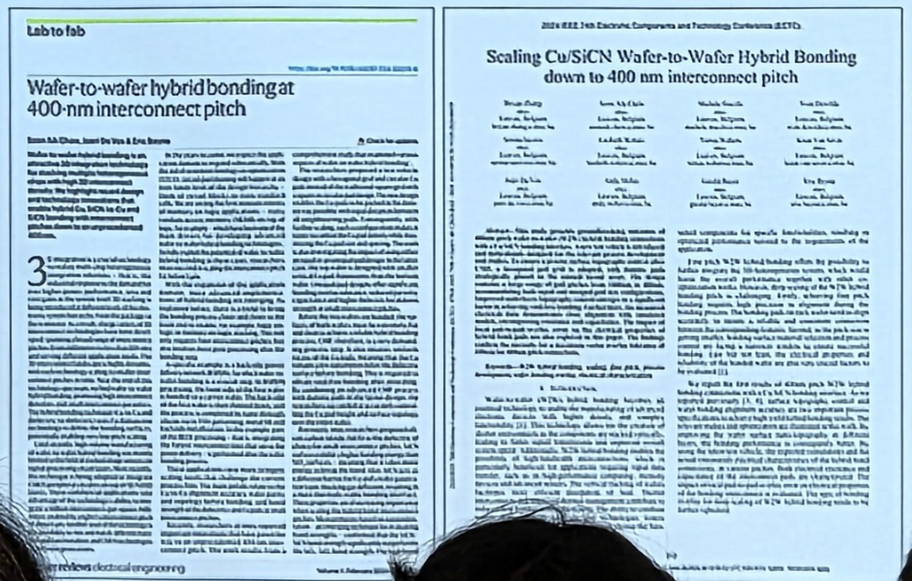
03. Interconnection

03 Interconnection

Interconnection : Fine-pitch and low-temperature bonding

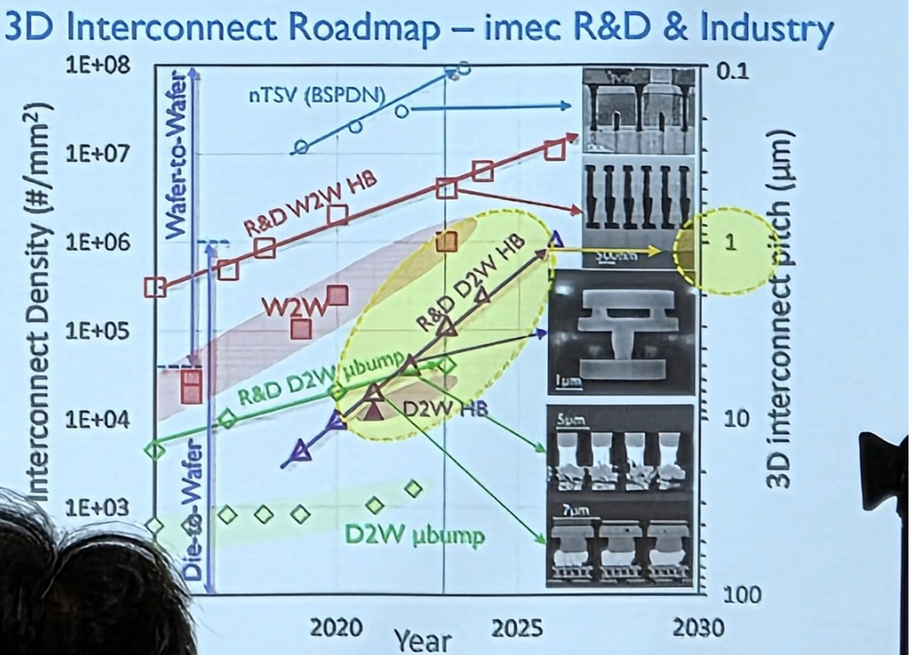
➢ Toward sub-micron pitch Cu/SiO_2 hybrid bonding

- Sub-micron pitch hybrid bonding (>1.0 μm) is mainly led by industry and major research centers such as IMEC
- As interconnect pitch decreases, Cu pad size, bonding gap, alignment tolerance, and contact resistance become critical
- This trend requires selective and damage-free interface engineering for both Cu and SiO_2 surfaces.


Lab to fab

Wafer-to-wafer hybrid bonding at 400-nm interconnect pitch

Scaling Cu/SiCN Wafer-to-Wafer Hybrid Bonding down to 400 nm interconnect pitch



【 EC roadmap of 3D interconnection 】

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03 Interconnection

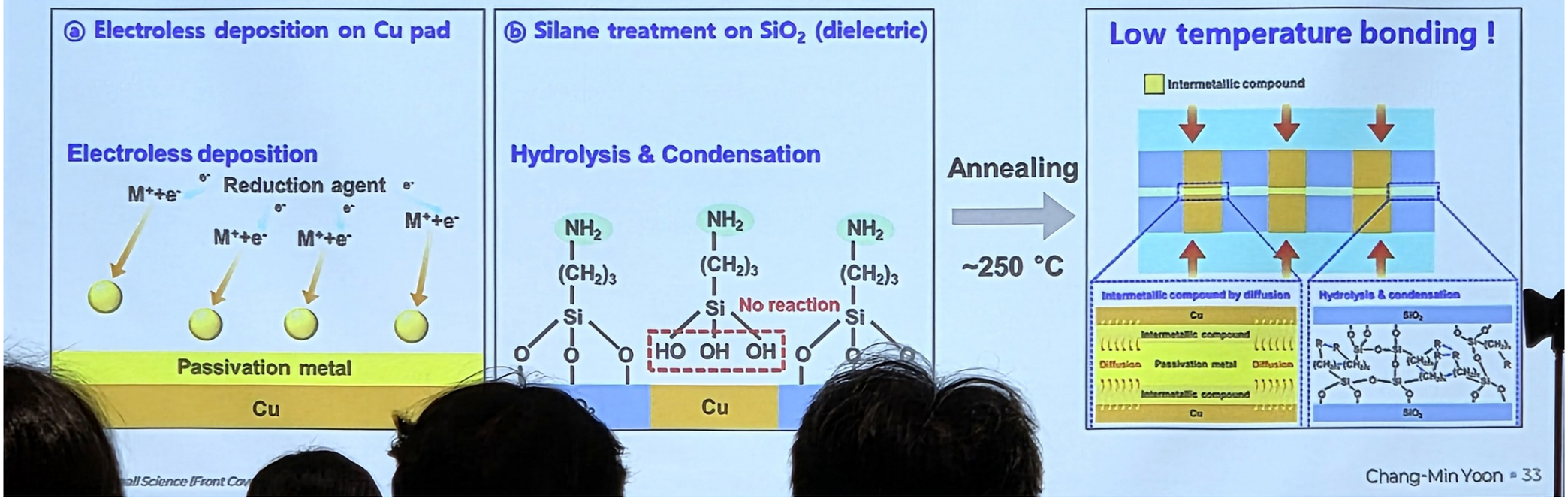
03 Interconnection

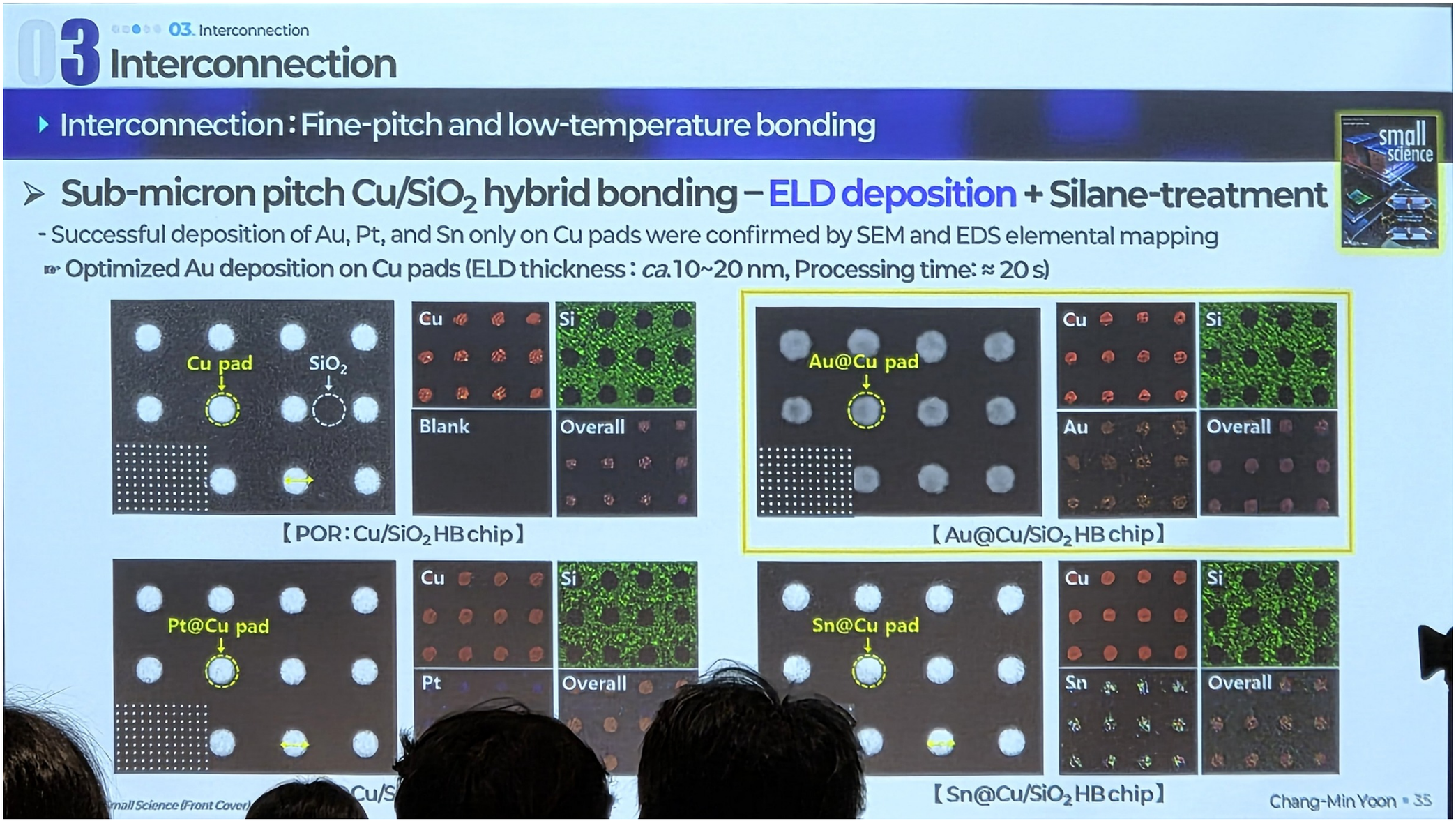
Interconnection : Fine-pitch and low-temperature bonding

➢ Sub-micron pitch Cu/SiO_2 hybrid bonding – ELD deposition + Silane-treatment

- Sub-micron pitch hybrid bonding requires damage-free and selective interface control to replace conventional method
- ⓐ Cu pad : Electroless metal deposition (ELD), ⓑ SiO_2 dielectric : Silane treatment

☞ Research target : Enabling low-temperature (250~300 °C) Cu/SiO_2 hybrid bonding *via* mild treatment method




03 Interconnection
03 Interconnection
Interconnection : Fine-pitch and low-temperature bonding
small science
Sub-micron pitch Cu/SiO2 hybrid bonding – ELD deposition + Silane-treatment
- Successful deposition of Au, Pt, and Sn only on Cu pads were confirmed by SEM and EDS elemental mapping
Optimized Au deposition on Cu pads (ELD thickness : ca.10~20 nm, Processing time: ≈ 20 s)
Cu pad
SiO2
Cu
Si
Blank
Overall
【POR : Cu/SiO2 HB chip】
Au@Cu pad
Cu
Si
Au
Overall
【Au@Cu/SiO2 HB chip】
Pt@Cu pad
Cu
Si
Pt
Overall
Sn@Cu pad
Cu
Si
Sn
Overall
【Sn@Cu/SiO2 HB chip】
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03 Interconnection

03 Interconnection

Interconnection : Fine-pitch and low-temperature bonding

➢ Sub-micron pitch Cu/SiO_2 hybrid bonding – ELD deposition + Silane-treatment

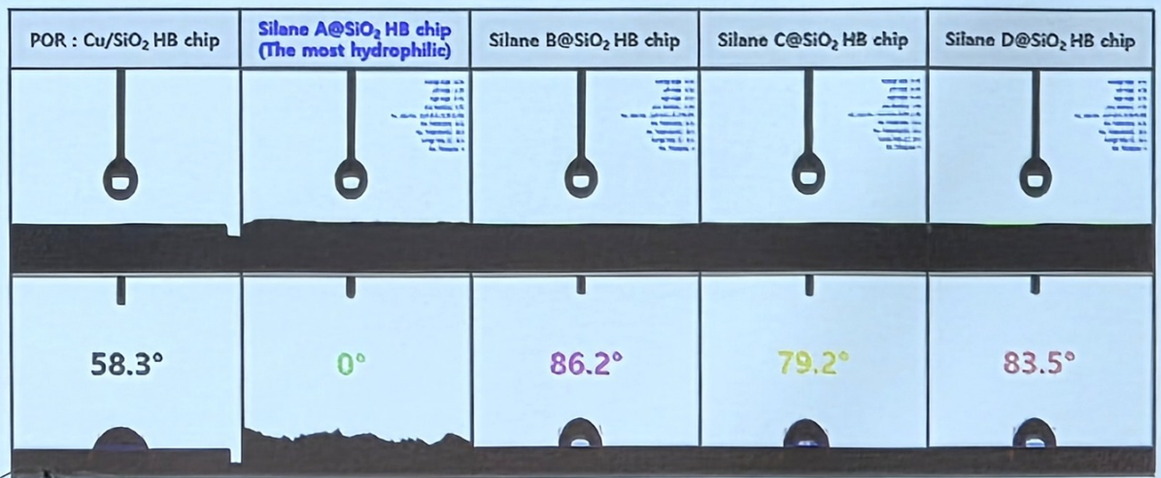
- 2.0 wt% silane solution was prepared using (3-aminopropyl)triethoxysilane (APS), octyltriethoxysilane (OTS), and hexadecyltriethoxysilane (HDS), and the solution was deposited by spin coating
- After the deposition of silane precursors onto the SiO_2 surface, the samples were dried

Step 1. Preparing spicement

Step 2. Silane coating (6000 rpm)

Step 3. Washing with EtOH

Step 4. Dry in



【 Silane deposition using s ace properties of silane-treated Cu/SiO_2 HB chip 】

mall Science (Front Cover)

Chang-Min Yoon • 36

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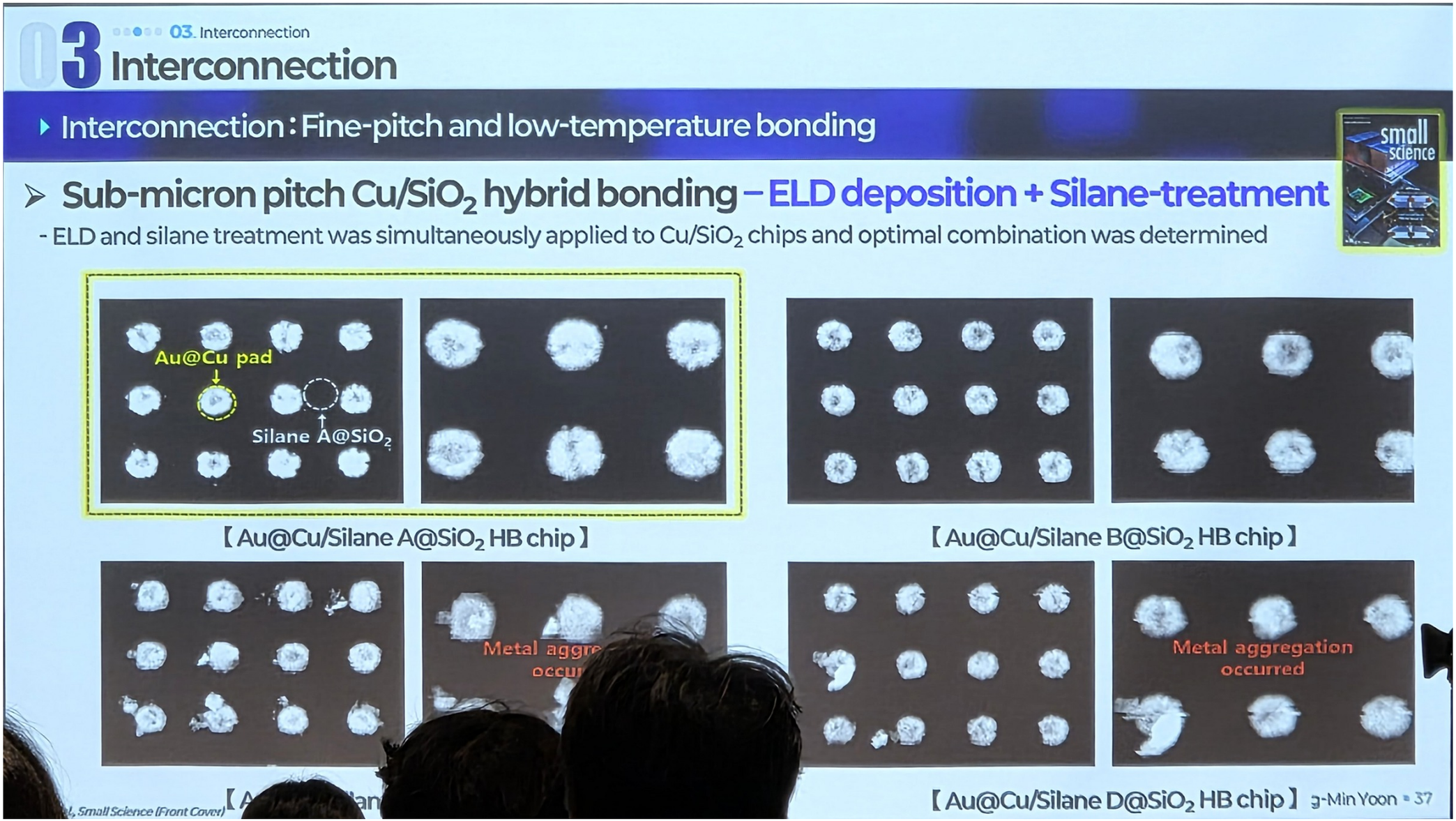

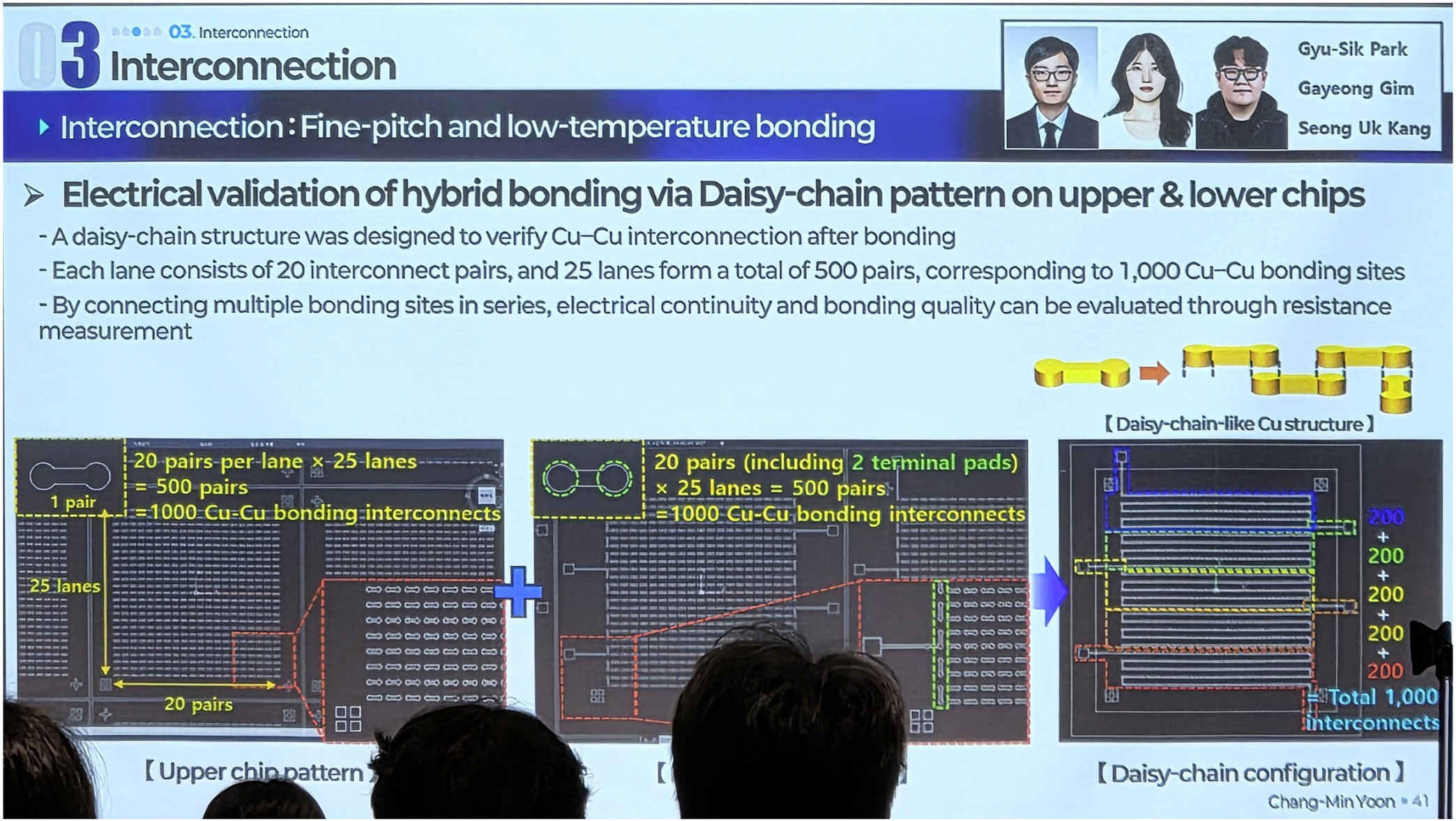
03 Interconnection
03. Interconnection
Interconnection : Fine-pitch and low-temperature bonding
Gyu-Sik Park
Gayeong Gim
Seong Uk Kang
Electrical validation of hybrid bonding via Daisy-chain pattern on upper & lower chips
- A daisy-chain structure was designed to verify Cu–Cu interconnection after bonding
- Each lane consists of 20 interconnect pairs, and 25 lanes form a total of 500 pairs, corresponding to 1,000 Cu–Cu bonding sites
- By connecting multiple bonding sites in series, electrical continuity and bonding quality can be evaluated through resistance measurement
【Daisy-chain-like Cu structure】
1 pair
20 pairs per lane × 25 lanes
= 500 pairs
=1000 Cu-Cu bonding interconnects
25 lanes
20 pairs
20 pairs (including 2 terminal pads)
× 25 lanes = 500 pairs
=1000 Cu-Cu bonding interconnects
200 + 200 + 200 + 200 + 200
= Total 1,000 interconnects
【Upper chip pattern】
【Daisy-chain configuration】
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03 Interconnection

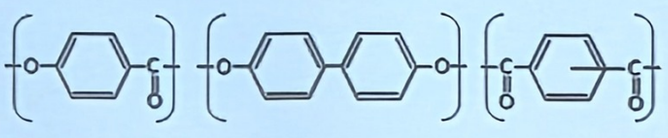
03 Interconnection

Interconnection : Fine-pitch and low-temperature bonding

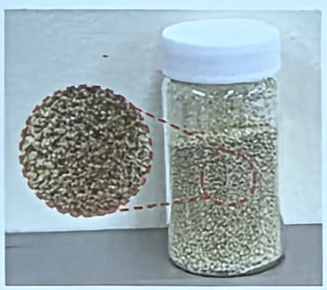
➢ Invention of new dielectric material : Liquid crystal polymer (LCP)

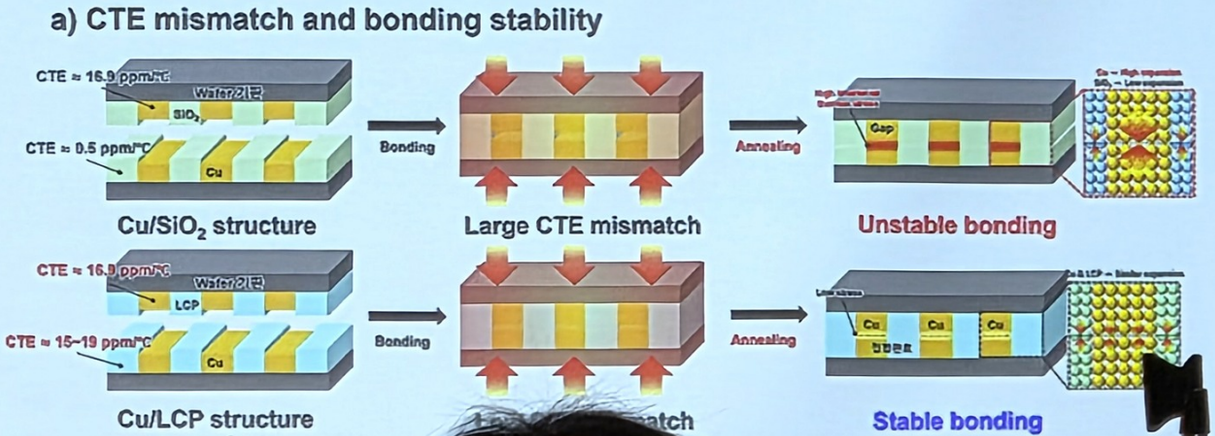
- To reduce the CTE differences, we have selected LCP with CTE of 15~19 ppm/°C as the dielectric materials (matching with Cu)
- Also, LCP low dielectric constant ($\varepsilon < 3$, lower than SiO_2), thus suitable as packaging dielectric materials
- However, the biggest challenge is utilizing thermoplastic LCP to substitute thermoset polymer for packaging application

☞ **Copolymerization of HBA, TPA, and BP → LCP (Tm ≈ 250–300 °C) → Film fabrication**



HBA
(*p*-Hydroxybenzoic acid)
TPA
(Terephthalic acid)
BP
(Biphenol)





【Chemical structure of L… 【Sch…erature hybrid bonding】

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04 Warpage and Thermal management

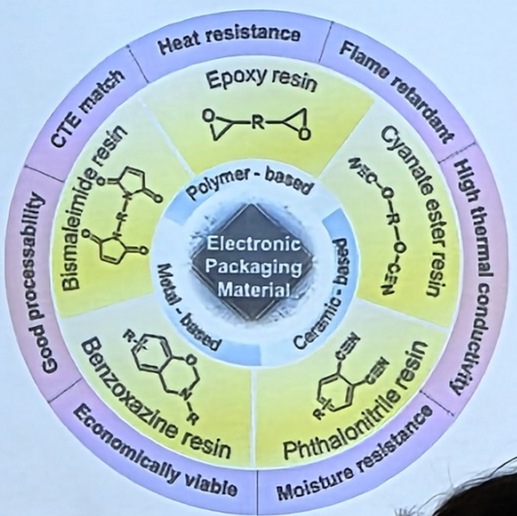
04 Warpage and Thermal management

▸ Warpage and Mechanical Reliability : Stress control in heterogenous stacks

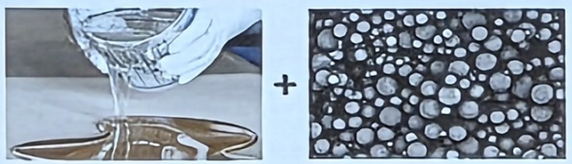
➢ Epoxy-based Materials as Key Building Blocks for Semiconductor Packaging

- Epoxy polymers are the most widely used thermosetting materials for electronics encapsulants
- Their excellent mechanical strength and thermal/chemical stability enable high-temperature processing and long-term reliability

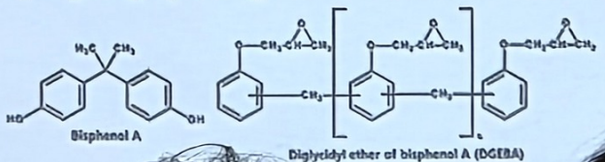
☞ By incorporating inorganic fillers such as SiO_2 into the epoxy matrix, various packaging materials with controlled CTE, thermal conductivity, and dielectric properties can be achieved

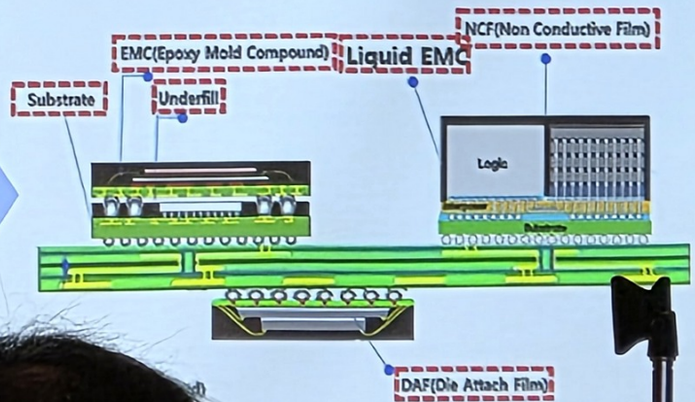


Epoxy resin + Silica filler



* Epoxy ring opening reaction mechanism





【 Epoxy resi d materials for packaging 】

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04 Warpage and Thermal management

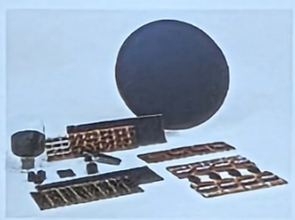
04 Warpage and Thermal management

Warpage and Mechanical Reliability : Stress control in heterogenous stacks

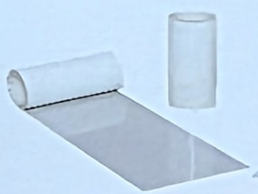
Functional Epoxy Materials for Advanced Semiconductor Packaging

- Epoxy resins are widely used as matrix materials in EMC, underfill, NCF, DAF, and package substrates
- Aromatic and aliphatic epoxy resins can be selected depending on targeted properties mechanical stability, viscosity, processability, and reliability requirements

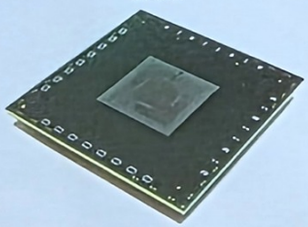
☞ Epoxy formulation design is a key materials strategy for stress control and heat dissipation in HBM and advanced packaging.



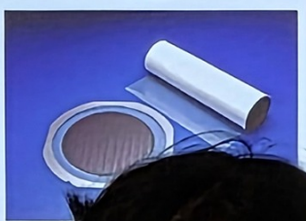
EMC

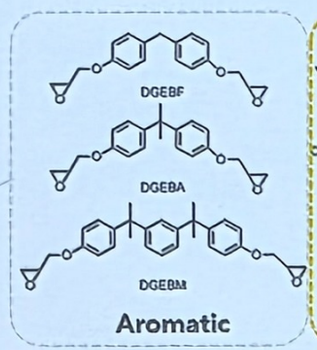


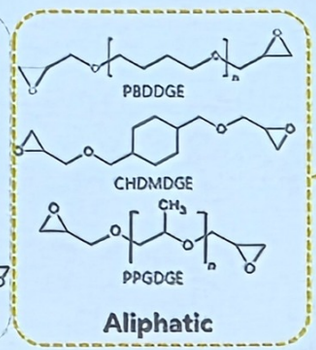
NCF



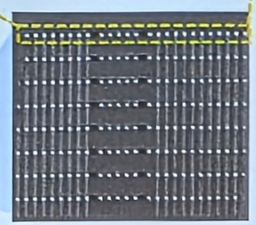
PCB

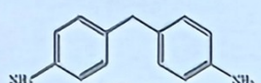


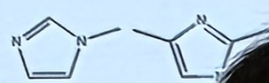




Underfill







agent

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04 Warpage and Thermal management

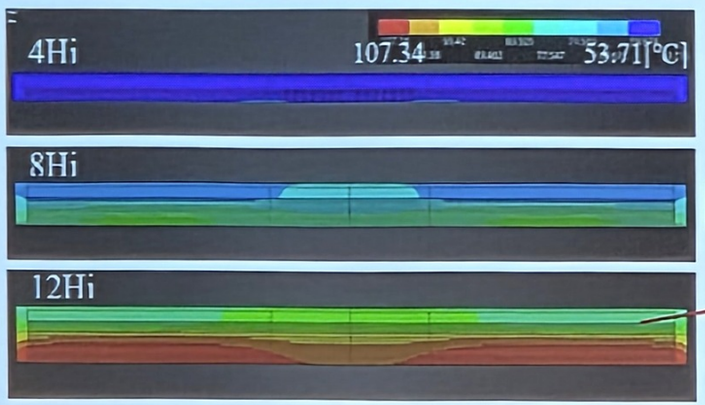
04 Warpage and Thermal management

▸ Warpage and Mechanical Reliability : Stress control in heterogenous stacks

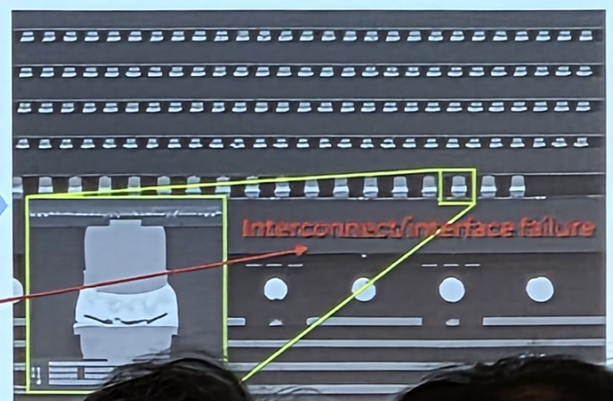
➢ Why Flexible Underfill Resin is Needed for High-stack HBM

- As HBM stack height increases, thermomechanical stress accumulates at chip-to-chip and interconnect interfaces
- CTE/modulus mismatch induces warpage, interfacial delamination, and interconnect failure
- Therefore, next-generation underfill should provide not only gap filling and adhesion, but also stress relaxation

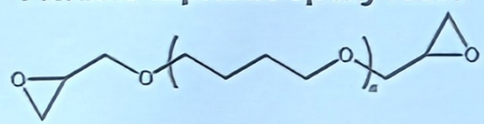
☞ **Flexible/rubber-like epoxy segments are required to suppress residual stress and improve mechanical reliability**



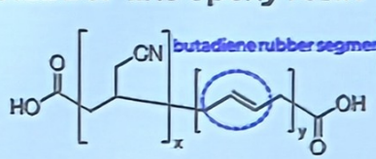
【Increasing thermal stres



* 4 Hi to 12 Hi HBM
- Flexible aliphatic epoxy resin



* Beyond12 Hi HBM
- Rubber-like epoxy resin



【Flexible & Rubber-like epoxy

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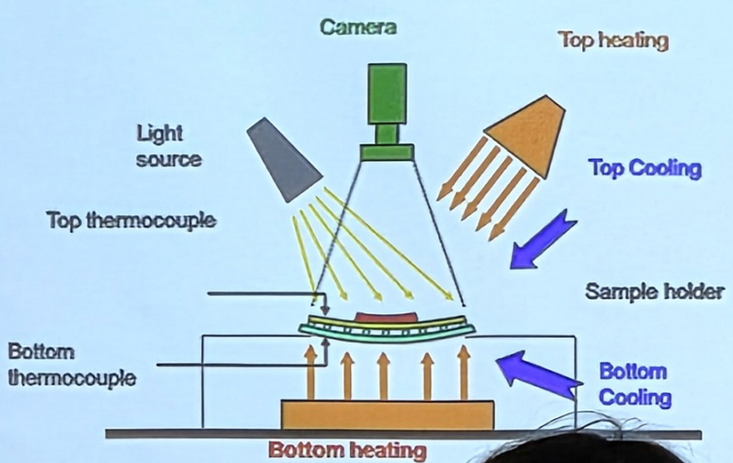
04 Warpage and Thermal management

04 Warpage and Thermal management

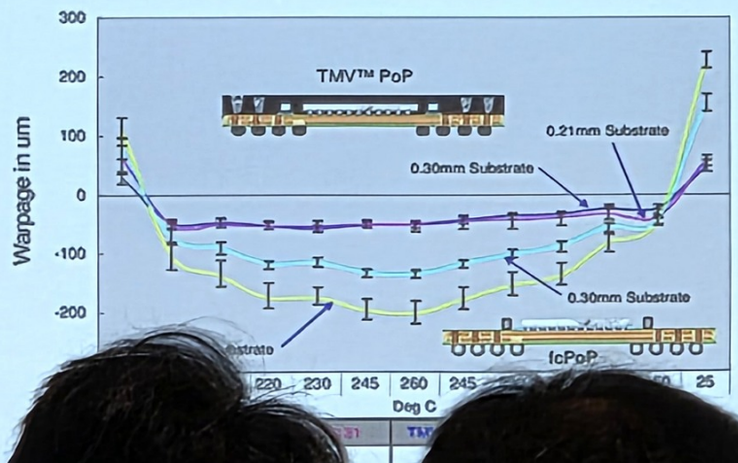
▸ Warpage and Mechanical Reliability : Stress control in heterogenous stacks

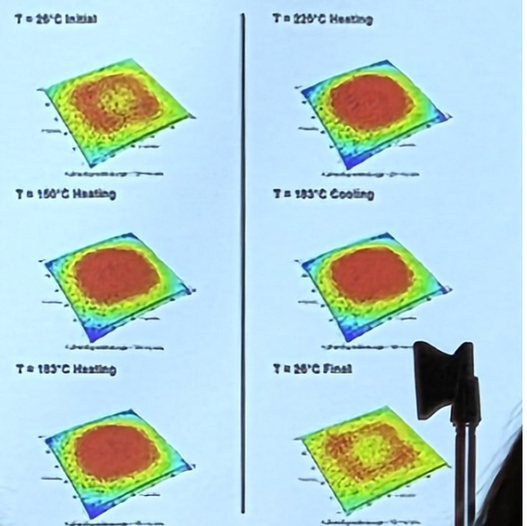
➢ Moiré analysis conditions

- The warpage behavior of the underfill-applied specimens was evaluated as a function of temperature using the Shadow Moiré method, including warpage–temperature plots and contour maps
- Temperature profile: initial temperature of 25 °C, maximum temperature of 260 °C, and final temperature of 25 °C
- Temperature steps: 25 → 50 → 75 → 100 → 125 → 150 → 165 → 175 → 200 → 215 → 245 → 260 → 245 → 215 → 200 → 175 → 165 → 150 → 125 → 100 → 75 → 50 → 25 °C



【Shadow moiré mea





【Contour mapping】

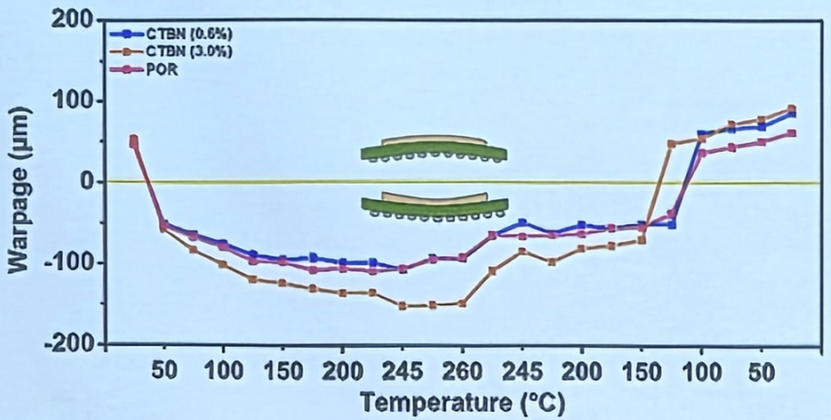
04 Warpage and Thermal management

04 Warpage and Thermal management

Warpage and Mechanical Reliability : Stress control in heterogenous stacks

➢ Moiré analysis results

- The warpage behavior was evaluated by calculating ① the variation range, ② the difference between the initial and final values, and ① + ② the total value
- Warpage behavior: POR > CTBN (0.6%) > CTBN (3.0%)
- CTBN (0.6%) showed a comparable level of warpage behavior to the POR sample



Sample	① Degree of deformation	② Recovery after deformation	① + ②
CTBN (0.6%)	144 μm	36 μm	180 μm
CTBN (3.0%)	202 μm	38 μm	240 μm
POR	140 μm	[illegible] μm	155 μm

【Warpage [illegible] mage】

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04 Warpage and Thermal management

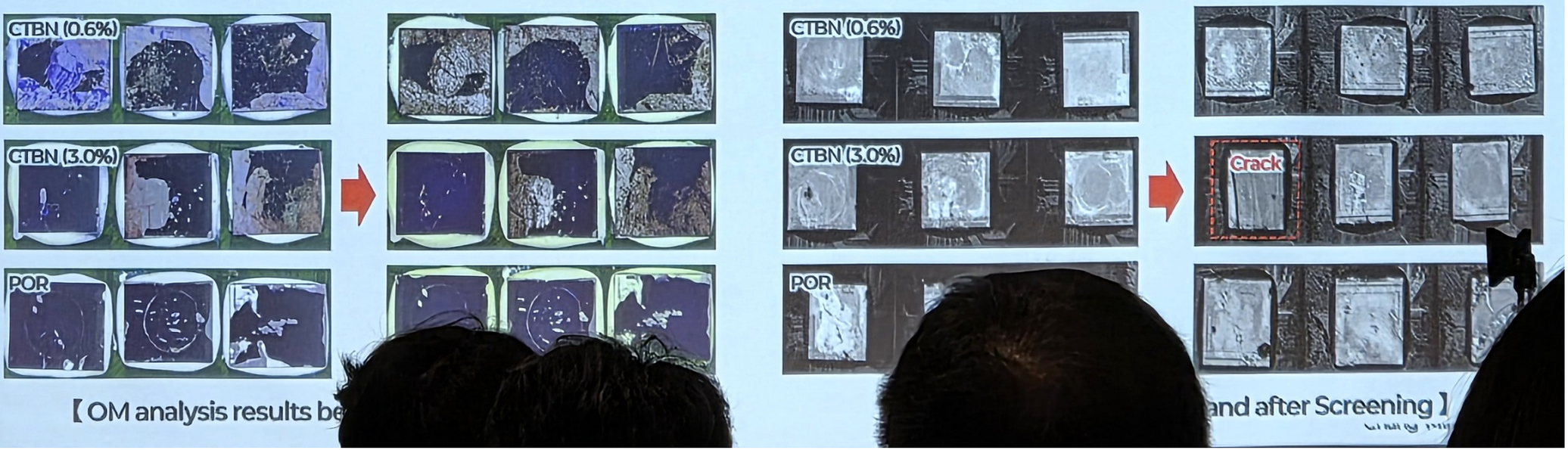
04 Warpage and Thermal management

▸ Warpage and Mechanical Reliability : Stress control in heterogenous stacks

➢ Screening test results

- After the screening test involving thermal aging and thermal shock, OM and SAT analyses were performed on the underfill samples
- Visual inspection of underfill: Yellowing of the underfill was observed
- Ultrasonic inspection of underfill: Chip cracking occurred in the CTBN (3.0%) sample

☛ This crack is presumed to be caused by reduced stress relaxation capability due to the high low-molecular-weight content



04 Warpage and Thermal management

04 Warpage and Thermal management

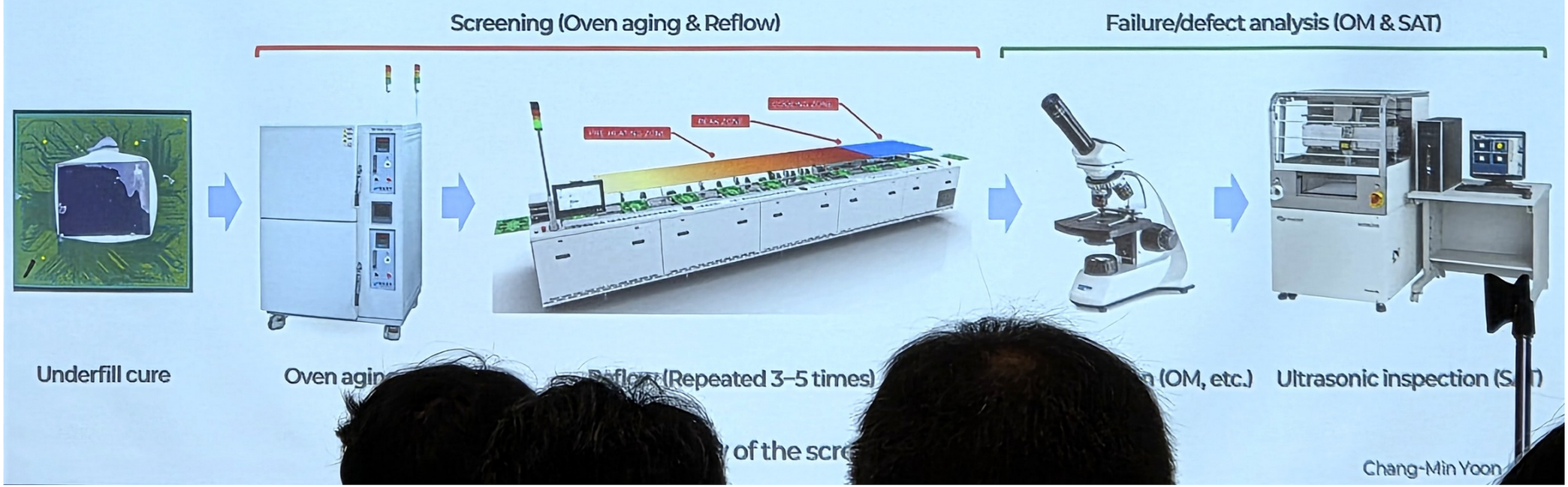
▸ Warpage and Mechanical Reliability : Stress control in heterogenous stacks

➢ Screening test for semiconductor packaging company A (Early failure analysis)

- Screening test: A reliability evaluation method used to detect defects by applying thermal aging and thermal shock to underfill
- Two screening steps were conducted, and the presence of failures or defects was evaluated using OM and SAT analysis

※ 1st Screening Condition: Underfill cure → Oven aging at 125 °C for 72 h → Multi-reflow at 260 °C for 4–8 min, repeated 3 times

※ 2nd Screening Condition: 1st screening → Oven aging at 150 °C for 48 h → Multi-reflow at 260 °C for 4–8 min, repeated 5 times



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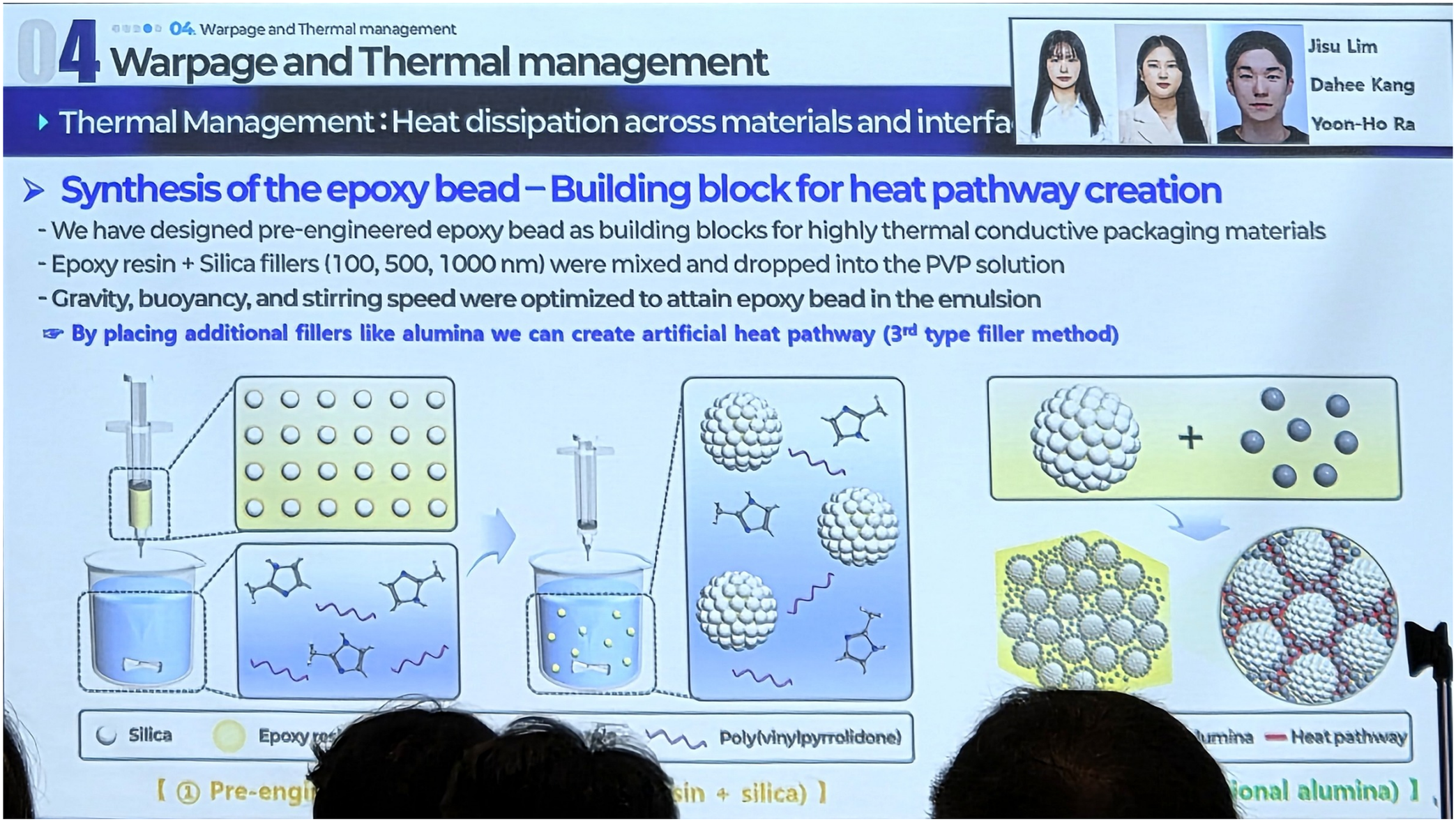

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– Thank you for your attention –

* Selected as a Winner of the 2026 Small Young Innovator Award !

Advanced Packaging & Polymeric Materials Laboratory

Prof. Chang–Min Yoon

- Dr. Latiful Kabir
- Suk Jekal (Ph. D course)
- Woohyeon Kim (Ph. D)
- Jisu Lim (Master course)
- Gyu–Sik Park (Master)
- Yoon–Ho Ra (Master)
- Dahee Kang (Master)
- Jaehyoung Park (Master)
- Gayeong Gim (Master)
- Minsong Jeon (Master)
- Seong Uk Kang (Master)
- Huiseong Eom (Master)
- Jae Yong [illegible]Master)

From 2027 March (new)

- 전윤영 (Ph. D)
- 최준범 (Master)
- 이재원 (Master)
- 김민서 (Master)
- 박진서 (Master)
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